Some Topics in Model Theory: Ultraproducts

UGP Presentation (MTH496)

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We denote this by $\mathcal{M} \equiv \mathcal{N}$. For example, $(\mathbb{R}, <) \equiv (\mathbb{Q}, <)$ and $\overline{\mathbb{Q}} \equiv \mathbb{C}$.

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Theorem

Let $\mathcal{M}, \mathcal{N}$ be saturated structures such that $\mathcal{M} \equiv \mathcal{N}$ and $|M| = |N|$. Then $\mathcal{M} \cong \mathcal{N}$.

This gives us an idea for a partial converse.

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4. For all $A \subseteq I$, either $A \in \mathcal{U}$ or $I \setminus A \in \mathcal{U}$.

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The following result appeared in JAMS (2016).

Theorem (Malliaris & Shelah, 2016)

$\mathfrak{p} = \mathfrak{t}$.

$$\mathfrak{p} = \min\{|\mathcal{F}| : \mathcal{F} \subseteq [\omega]^\omega \text{ has s.f.i.p but no pseudo-intersection}\}$$
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This long standing open problem from general topology/set theory was connected to the model theoretic question: **Is $\mathbf{SOP_2}$ maximal in Keisler's order?**
The notion of (good regular) ultrafilters is central to the above result and in fact, a new characterization of such ultrafilters.

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This new structure, denoted by $\prod_{\mathcal{U}} \mathcal{M}_i$, is the **ultraproduct** of $\langle \mathcal{M}_i : i \in I \rangle$ over $\mathcal{U}$.

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- Recall: we wanted to find a bigger structure $\mathcal{M}^*$ such that $\mathcal{M} \equiv \mathcal{M}^*$.
- Łoś' theorem $\implies \mathcal{M} \equiv \prod_{\mathcal{U}} \mathcal{M}$.
- Want: $\prod_{\mathcal{U}} \mathcal{M}$ is saturated and $|\prod_{\mathcal{U}} \mathcal{M}| = |\prod_{\mathcal{U}} \mathcal{N}|$

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Suppose such a funtion h exists. Then $\Gamma(x)$ is realized in $\mathcal{M}^*$.

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Definition (Good ultrafilter)

An ultrafilter $\mathcal{U}$ is κ^+-good if for every type $\Gamma(x)$ over $\leq \kappa$ parameters, a function $h : [\Gamma]^{<\omega} \to \mathcal{U}$ satisfying (1) + (2) + (3) exists.

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Theorem (Keisler 1961 (GCH), Shelah 1971 (ZFC))

$\mathcal{M} \equiv \mathcal{N} \iff$ *there is an ultrafilter* $\mathcal{U}$, *such that* $\prod_{\mathcal{U}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{N}$.

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- $|\prod_{\mathcal{U}} \mathcal{M}| = |\prod_{\mathcal{U}} \mathcal{N}| = 2^\kappa$.
- By GCH, $2^\kappa = \kappa^+$. Uniqueness of saturated models gives, $\prod_{\mathcal{U}} \mathcal{M} \cong \prod_{\mathcal{U}} \mathcal{N}$. $\square$

Thank You!

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questions?